

14] Extendible Hashing: (Example - Photos) (ADS Notebook)

- Extendible Hashing is a dynamic hashing method wherein directories and buckets are used to hash data.
- The directories store address of the buckets in pointers.
- The buckets are used to hash the actual data.

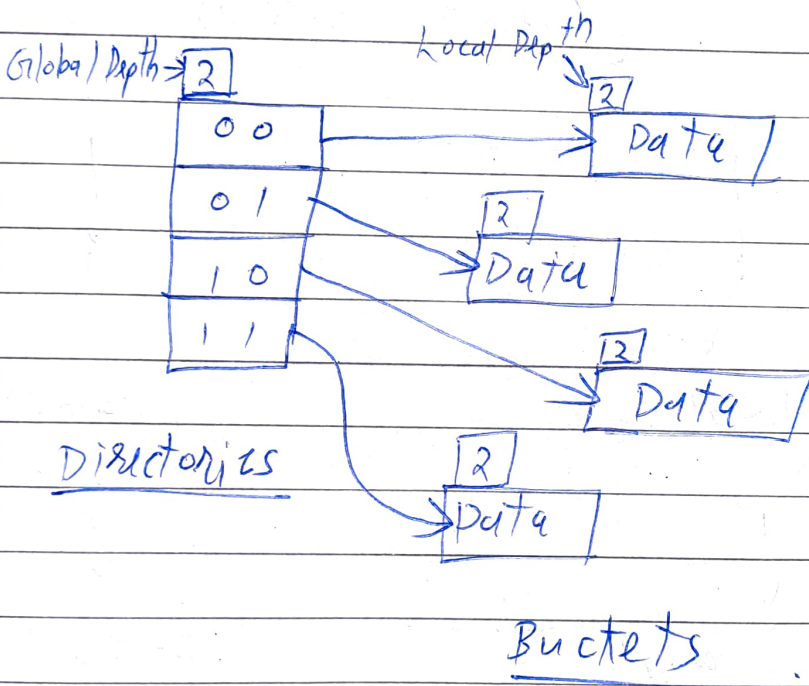


Fig. Basic structure of Extendible Hashing

- Number of directories = $2^{\text{Global Depth}}$

associated with directories

Page No. 18

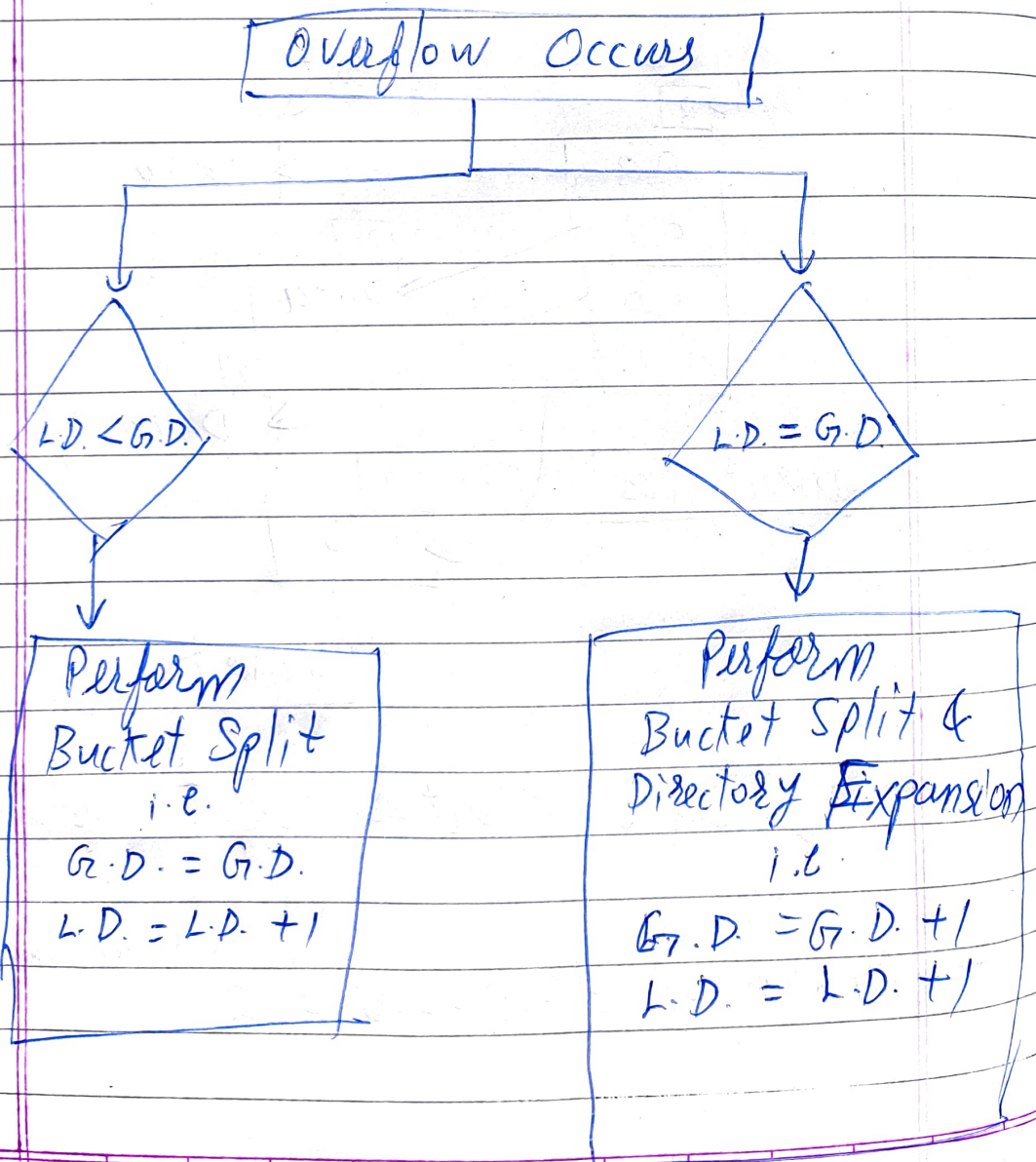
Date

→ Global depth = Number of bits in directory id.

→ Local depth = Associated with buckets.

$$L.D. \leq G.D.$$

→ Bucket size : How many elements to store in one bucket.



16 8 4 2 1

16 — 1 0 0 0 0

4 — 0 0 1 0 0

6 — 0 0 1 1 0

22 — 1 0 1 1 0

24 — 1 1 0 0 0

10 — 0 1 0 1 0

31 — 1 1 1 1 1

7 — 0 0 1 1 1

9 — 0 1 0 0 1

20 — 1 0 1 0 0

26 — 1 1 0 1 0

Insert

* Step 1 :- 16, 4, 6

11
0
1

11
16, 4, 6
0
1

→ Insert 32 (overflow)

G+1

L+1

2
00
01
10
11

2
16, 4, 24
00

2
6, 22, 10
10

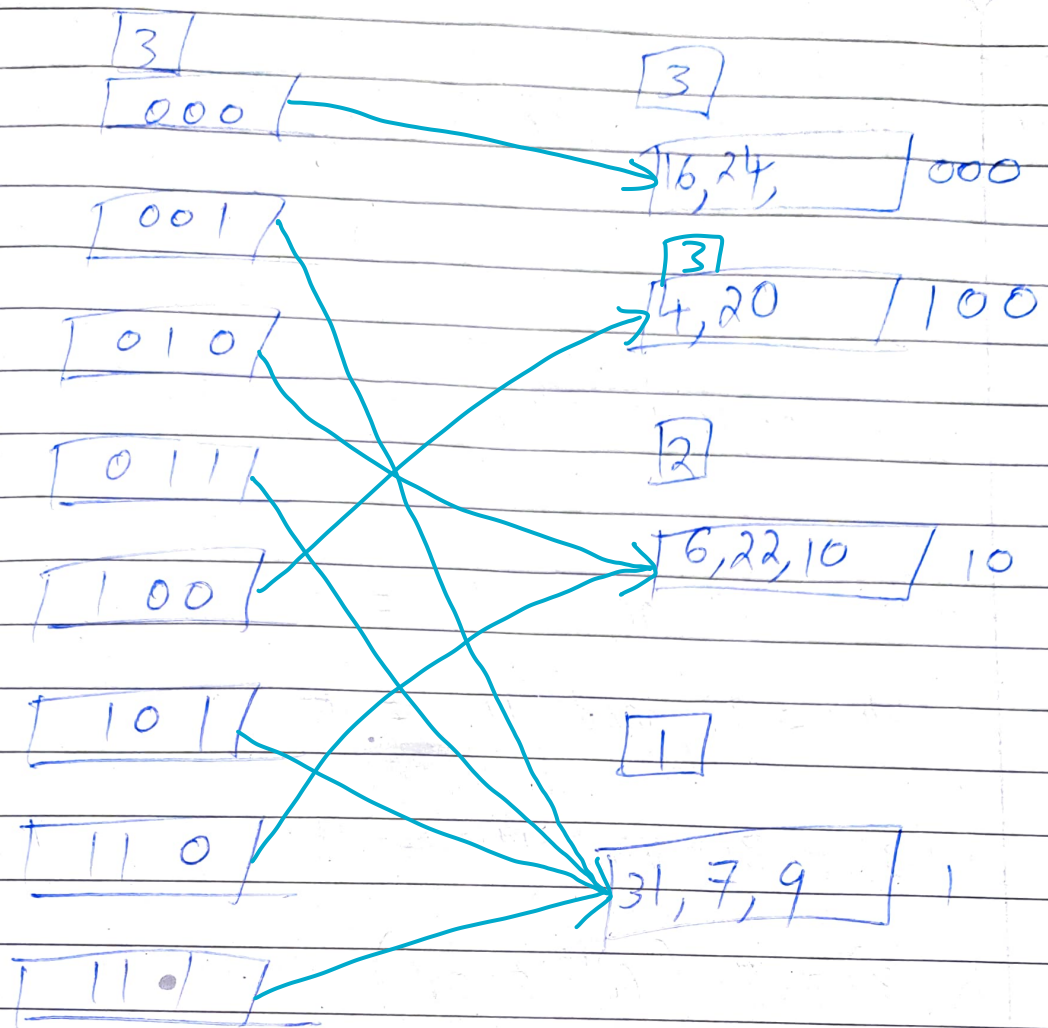
11
31, 7, 9
1

Now, insert 24, 10, 31, 7, 9

→ Insert 20 (overflow)

G+1

L+1



Insert 26 (Overflow)

But, $L < G$

$L+1$

